[BUG] [Return-types] Update the output shape of `qml.probs` for `default.q

https://github.com/PennyLaneAI/pennylane/issues/1607

ROW 100

[BUG] [Return-types] Update the output shape of `qml.probs` for `default.qubit.torch` #1607

New issue

Closed

antalszava opened on Aug 30, 2021

Expected behavior

Using `default.qubit.torch` and backpropagation, `return qml.probs(0), qml.probs(1)` returns a `(2, 2)` shape tensor (just as with TensorFlow).

Actual behavior

A `(4,)` tensor is returned for `return qml.probs(0), qml.probs(1)` in Torch+backprop. This could be due to differing behaviour in the `DefaultQubitTorch._asarray()` method.

Additional information

No response

Source code

```
import pennylane as qml

dev = qml.device('default.qubit.torch', wires=3)

@qml.qnode(dev, interface='torch')
def circuit():
    return qml.probs(0), qml.probs(1)

circuit()
```

```
tensor([1., 0., 1., 0.], dtype=torch.float64)
```

```
import pennylane as qml

dev = qml.device('default.qubit.tf', wires=3)

@qml.qnode(dev, interface='tf')
def circuit():
    return qml.probs(0), qml.probs(1)

circuit()
```

```
<tf.Tensor: shape=(2, 2), dtype=float64, numpy=
array([[1., 0.],
       [1., 0.]])>
```

```


### Tracebacks

_No response_

### System information

```shell
Name: PennyLane
Version: 0.18.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/XanaduAI/pennylane
Author: None
Author-email: None
License: Apache License 2.0
Location: /xanadu/pennylane
Requires: numpy, scipy, networkx, autograd, toml, appdirs, semantic-version, autoray, cachetools
Required-by: PennyLane-Cirq, PennyLane-Orquestra, PennyLane-Qchem, PennyLane-SF, pennylane-qulacs, PennyLane-]
Platform info:           Linux-5.11.0-27-generic-x86_64-with-glibc2.10
Python version:          3.8.5
```

Assignees
antalszava
rmoyard

Labels
bug 🐛

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

Development
No branches or pull requests

Notifications Customize
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Participants
```

```
- cirq.qsimh (PennyLane-Cirq-0.17.1)
- cirq.simulator (PennyLane-Cirq-0.17.1)
- orquestra.forest (PennyLane-Orquestra-0.15.0)
- orquestra.ibmq (PennyLane-Orquestra-0.15.0)
- orquestra.qiskit (PennyLane-Orquestra-0.15.0)
- orquestra.qulacs (PennyLane-Orquestra-0.15.0)
- strawberryfields.fock (PennyLane-SF-0.16.0.dev0)
- strawberryfields.gaussian (PennyLane-SF-0.16.0.dev0)
- strawberryfields.gbs (PennyLane-SF-0.16.0.dev0)
- strawberryfields.remote (PennyLane-SF-0.16.0.dev0)
- strawberryfields.tf (PennyLane-SF-0.16.0.dev0)
- qulacs.simulator (pennylane-qulacs-0.17.0.dev0)
- ionq.qpu (PennyLane-IonQ-0.17.0.dev0)
- ionq.simulator (PennyLane-IonQ-0.17.0.dev0)
- braket.aws.qubit (amazon-braket-pennylane-plugin-1.4.1.dev0)
- braket.local.qubit (amazon-braket-pennylane-plugin-1.4.1.dev0)
- forest.numpy_wavefunction (PennyLane-Forest-0.17.0.dev0)
- forest.qvm (PennyLane-Forest-0.17.0.dev0)
- forest.wavefunction (PennyLane-Forest-0.17.0.dev0)
- honeywell.hqs (PennyLane-Honeywell-0.16.0.dev0)
- qiskit.aer (PennyLane-qiskit-0.18.0.dev0)
- qiskit.basicaer (PennyLane-qiskit-0.18.0.dev0)
- qiskit.ibmq (PennyLane-qiskit-0.18.0.dev0)
- lightning.qubit (PennyLane-Lightning-0.18.0.dev0)
- default.gaussian (PennyLane-0.18.0.dev0)
- default.mixed (PennyLane-0.18.0.dev0)
- default.qubit (PennyLane-0.18.0.dev0)
- default.qubit.autograd (PennyLane-0.18.0.dev0)
- default.qubit.jax (PennyLane-0.18.0.dev0)
- default.qubit.tf (PennyLane-0.18.0.dev0)
- default.qubit.torch (PennyLane-0.18.0.dev0)
- default.tensor (PennyLane-0.18.0.dev0)
- default.tensor.tf (PennyLane-0.18.0.dev0)
```

- [x] I have searched exisiting GitHub issues to make sure the issue does not already exist.

antalszava added bug on Aug 30, 2021

github-actions assigned antalszava on Aug 30, 2021

antalszava mentioned this on Aug 30, 2021

Update tests to ensure that Torch backpropagation is supported #1598

josh146 mentioned this on Sep 21, 2021

Add autodiff integration tests for the new QNode pipeline #1646

antalszava mentioned this on Jul 29, 2022

Use parameter broadcasting in `param_shift` #2749

dwierichs on Aug 2, 2022 Contributor ···

An instance of where this bug became apparent/relevant is the use of broadcasting in `qml.gradients.param_shift`: Due to the differing output shape of the torch device in its `_asarray` method, the if-condition in `gradients/parameter_shift._evaluate_gradient` needs to check for torch devices and handles the corresponding case differently than for all other interfaces. This leads to self-consistent behaviour within the torch interface/device scenario, but showcases that the `_asarray` method should be adapted to yield consistent results with other devices or produce predictable output shapes instead of iteratively flattening and concatenating the returned values.

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Jaybsoni assigned rmoyard on Aug 2, 2022

AlbertMitjans assigned AlbertMitjans and unassigned AlbertMitjans on Aug 10, 2022

Jaybsoni assigned rmoyard on Aug 2, 2022

AlbertMitjans assigned AlbertMitjans and unassigned AlbertMitjans on Aug 10, 2022

antalszava on Nov 29, 2022 — Contributor — Author

Note that this bug will no longer be relevant with the new return types:

In [1]:
```python
import pennylane as qml

qml.enable_return()

dev = qml.device('default.qubit.torch', wires=3)

@qml.qnode(dev, interface='torch')
def circuit():
    return qml.probs(0), qml.probs(1)

circuit()
```

Out[1]:
```
(tensor([1., 0.], dtype=torch.float64), tensor([1., 0.], dtype=torch.float64))
```

In [2]:
```python
import pennylane as qml

qml.enable_return()

dev = qml.device('default.qubit.tf', wires=3)

@qml.qnode(dev, interface='tf')
def circuit():
    return qml.probs(0), qml.probs(1)

circuit()
```

Out[2]:
```
(<tf.Tensor: shape=(2,), dtype=float64, numpy=array([1., 0.])>,
 <tf.Tensor: shape=(2,), dtype=float64, numpy=array([1., 0.])>)
```

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antalszava changed the title ~~[BUG] Update the output shape of `qml.probs` for `default.qubit.torch`~~ [BUG] [Return types] Update the output shape of `qml.probs` for `default.qubit.torch` on Nov 29, 2022

antalszava changed the title ~~[BUG] [Return types] Update the output shape of `qml.probs` for `default.qubit.torch`~~ [BUG] [Return-types] Update the output shape of `qml.probs` for `default.qubit.torch` on Nov 29, 2022

albi3ro on Apr 4, 2023 — Contributor

Given this bug is fixed by the new return types specification, I'm marking this bug as done.

albi3ro closed this as completed on Apr 4, 2023


Use parameter broadcasting in param_shift #2749
Code
Merged
dwierichs merged 184 commits into master from parameter-broadcasting-4 on Aug 3, 2022
Conversation 82
Commits 184
Checks 0
Files changed 5
+1,400 −191
dwierichs commented on Jun 20, 2022
Contributor
This PR follows up on #2627 and makes use of the newly introduced parameter broadcasting in qml.gradients.param_shift .
Due to inconsistent output handling, it currently does not support tapes/QNodes with multiple return values and does not support
shot vectors yet.
Reviewers
antalszava
mlxd
Assignees
No one assigned
Labels